

XI DING

xding232@wisc.edu | [Google Scholar](#) | [Personal Website](#)

SUMMARY

Xi Ding is a Ph.D. student in Computer Sciences at the University of Wisconsin–Madison. Prior to his Ph.D., he received a Master of Computing (Machine Learning) from the Australian National University in 2025, and conducted research at Carnegie Mellon University and the ARC Research Hub at Griffith University. His research focuses on AI/ML, spanning test-time scaling of LLMs and VLMs, trustworthiness and interpretability of AI, and multimodal model learning. Motivated by designing AI systems that not only perform well but also align with human understanding, operate transparently, and remain robust in high-stakes settings, he has published at ICLR, AACL, EMNLP, and WWW workshop, where his work received a Best Paper Award and an Oral presentation.

RESEARCH INTERESTS

Machine Learning, Representation Learning, Multimodal Models, LLMs & AI Agents, Trustworthy AI & Interpretability, Video Understanding

EDUCATION

University of Wisconsin–Madison <i>Ph.D. in Computer Sciences</i>	Aug. 2026 - Present
Australian National University <i>Master of Computing (Machine Learning)</i> GPA (WES converted): 3.85/4.0	July 2023 - July 2025
China Agricultural University & University of Colorado Denver <i>Bachelor of Arts (Economics)</i> GPA: 3.37/4.0	Sep. 2019 - June 2023

PUBLICATIONS

Peer-Reviewed Publications

- [1] Y. Hu, X. Tong, H. Bai, **X. Ding**, S. M. Bharadwaj, S. Cao, R. Nowak, and J. Zhang, “Understanding why language models hallucinate: Testing reasoning against priors,” in *Findings of the Association for Computational Linguistics: EMNLP*, 2026.
- [2] L. Wang*, **X. Ding***, Y. Gao, and P. Koniusz, “Subspace kernel learning on tensor sequences,” in *International Conference on Learning Representations (ICLR)*, 2026.
- [3] **X. Ding**, L. Wang, P. Koniusz, and Y. Gao, “Learning time in static classifiers,” in *AAAI Conference on Artificial Intelligence (AAAI)*, 2026.
- [4] **X. Ding** and L. Wang, “The journey of action recognition,” in *Companion Proceedings of the ACM Web Conference 2025*, May 2025, pp. 1869–1884. **(Best Paper Award)**
- [5] **X. Ding** and L. Wang, “Do language models understand time?,” in *Companion Proceedings of the ACM Web Conference 2025*, May 2025, pp. 1855–1868. **(Oral)**

Manuscripts Under Review

- [6] **X. Ding**, L. Wang, S.-H. Li, and Y. Gao, “Trust-aware joint feature-prediction discrepancy for robust domain adaptation,” under review at *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*.
- [7] Q. Trinh*, **X. Ding***, Y. Liu*, Z. Qin, X. Li, G. Durak, H. E. Aktas, E. Keles, U. Bagci, and M. Xu, “Beyond medical diagnostics: How medical multimodal large language models think in space,” under review at the *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*.

Preprints

- [8] **X. Ding** and L. Wang, “Quo vadis, anomaly detection? LLMs and VLMs in the spotlight,” arXiv preprint, 2024.

RESEARCH EXPERIENCE

Xu Lab, Carnegie Mellon University, Pittsburgh, USA

Oct. 2025 - May 2026

Research Intern to Prof. Min Xu

Conducting research on explainable machine learning, with a focus on language models and vision-language models.

ARC Research Hub, Griffith University, Brisbane, Australia

Mar. 2025 - Nov. 2025

ARC Hub Scholar (TIME@ARC Hub and Griffith) to Prof. Yongsheng Gao

Researched interpretable and explainable machine learning to advance transparent, robust, and trustworthy AI. Developed trust-aware domain adaptation, temporal reasoning for static classifiers, and uncertainty-aware kernel methods for tensor data.

TIME Lab, Australian National University, Canberra, Australia

June 2024 - Mar. 2025

Research Assistant to Dr. Lei Wang

Focused on advancing video understanding through large language models, with work spanning temporal reasoning, anomaly detection, and action recognition. Explored methods to capture long-range dependencies, improve interpretability, and enhance generalization in dynamic, open-world video tasks.

SELECTED RESEARCH PROJECTS

TIME Space

Feb 2025 - Nov. 2025

Client of The TechLauncher Project, ANU

Developed an interactive web platform for TIME Lab's research showcase, featuring ML-based feature extraction and embedding-driven retrieval for automated image tagging, clustering, and semantic search across temporal, spatial, and identity dimensions. Delivered a scalable, data-driven system through full-stack development and cloud deployment.

Active Vision

July 2024 - June. 2025

Leader of The TechLauncher Project, ANU

Developed a machine learning-powered computer vision system for real-time tennis swing analysis. High-speed cameras capture motion data, which is processed to extract key performance indicators such as swing speed, angle. The system provides instant feedback via an intuitive interface, aiding performance assessment and technique refinement.

Image Embedding Explorer

Aug. 2024 - Nov. 2024

Student of The Advanced Deep Learning Course Project, ANU

Built an interactive tool for visualizing high-dimensional image embeddings using projection pursuit. Enables intuitive exploration of embedding structures, linking each data point to its original image for enhanced interpretability in advanced deep learning models, such as CLIP and DINO.

TEACHING & ACADEMIC SERVICE

Teaching Assistant, COMP SCI/ECE 760 Machine Learning

Fall 2026

University of Wisconsin-Madison

Invited Speaker, Data61/CSIRO ICVG Reading Group

Jul. 2025

Delivered a talk titled "Echoes in the Model: When Features Reflect Predictions."

Workshop Coordinator, The Web Conference (WWW) 2025

Nov. 2024 - Apr. 2025

Coordinated the TIME 2025 workshop at WWW 2025, organizing speaker invitations, managing paper reviews, and facilitating cross-domain knowledge exchange.

SELECTED AWARDS & HONORS

Best Paper Award, The Web Conference (WWW 2025) Workshop

Apr 29, 2025

TIME 2025 Service Medal, TIME Lab

May 1, 2025

Outstanding Academic Achievements Scholarship, University of Colorado Denver

Dec 1, 2021